

Serena Huang

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EDUCATION

Massachusetts Institute of Technology

B.S., Double Major in Mathematics and Artificial Intelligence & Decision Making

Cambridge, MA

Expected May 2029

Activities: HackMIT Development Team, Software Staff for Harvard-MIT Mathematics Tournament & MIT Informatics Tournament

PROFESSIONAL EXPERIENCE

Cybersecurity at MIT Sloan School of Management (CAMS)

Cambridge, MA

Undergraduate Research Assistant

Sep 2026 - Present

- Evaluate phishing generation and detection across 8 open-weight and frontier LLMs, studying effects of model size and agentic scaffolding and implementing generator-critic refinement loops using Hugging Face Transformers and 4-bit quantization

Jane Street

New York, NY

Scholar, Academy of Math & Programming (AMP)

Jun 2026 - Jul 2026

- Completed 200+ hours of advanced coursework in combinatorics, probability, number theory, and data structures and algorithms taught by IMO medalists, and attended industry expert talks on machine learning and financial engineering
- Placed top 5 of 78 in algorithmic trading competition by implementing reinforcement learning and rule-based trading strategies

Bear Studios Consulting Group

St. Louis, MO

Technology Consultant

Dec 2023 - Apr 2026

- Developed digital placemaking mobile app for 50+ local businesses using React Native, Express.js, SQL, and AWS, and integrated map APIs, live video stream, and sensor data to enhance real-time user experience
- Overhauled website using Figma and React.js for celebrity gift bag client featuring 50+ brand partners and 100+ A-list events with 30M+ viewers, including the Grammys, Emmys, and Dancing with the Stars

Stanford University

Stanford, CA (Remote)

Section Leader, Code in Place Program

Apr 2025 - Jun 2025

- Selected from competitive global applicant pool to teach Python through Stanford's CS106A course to 40K+ learners
- Mentored 15 students in debugging and problem-solving during weekly discussion sections, and provided detailed code feedback

NASA & UT Austin's Center for Space Research

Austin, TX

Aerospace Research Intern, STEM Enhancement in Earth Science (SEES) Program

May 2024 - Aug 2024

- Applied k-means clustering and cosine similarity using NumPy and Scikit-Learn to compare 50+ terrestrial building materials, identifying Portland cement and brick as 93.5% matches to lunar basalt and regolith for Artemis mission planning
- Analyzed NASA astrophysical datasets to catalog 7.9K+ stars and generate velocity dispersion and temperature trend maps
- Invited to present findings at American Geophysical Union's 2024 Annual Fall Meeting through Bright STaRs program

Ariel Premium Supply

St. Louis, MO

Software Engineer Intern

Jun 2022 - Aug 2022 & Jun 2023 - Aug 2023

- Applied transfer learning with EfficientNet in PyTorch to learn visually similarity vector embeddings of over 10K+ images collected from U.S. Patent & Trademark Office, detecting near-duplicate logos to flag potential trademark infringement
- Integrated vector database into production logo search pipeline with OCR with 85% retrieval success rate on validation data
- Built Selenium and Linux cron data pipeline scrape 600+ client records daily, saving 15+ hours per week and \$12K+ annually

PERSONAL PROJECTS

AI Hydrological Modeling Research | PyTorch, Pandas, Matplotlib

github.com/serenahuang225/transformer-hydro-model

- Developed transformer-based rainfall single-basin and multi-basin models with PyTorch on CAMELS dataset over 531 basins with 20-40 years of vegetation, topography, and soil data, achieving Nash-Sutcliffe Efficiency score of 0.764 and outperforming LSTM baselines by 6%, classified as "Very Good" by hydrological modeling standards

Tunas Swim Meet Dashboard | Next.js, FastAPI, MCMC

github.com/serenahuang225/tunas-webapp-frontend

- Developed FastAPI backend hosted on Railway with Pydantic models and Markov Chain Monte Carlo (MCMC) simulator to analyze 800K+ race and relay outcomes for 1.2K+ USA Swimming clubs and 26K+ swimmers
- Built Next.js, TypeScript, and Recharts dashboard to visualize swimmer history, analytics, and relay optimization

Lively: All-in-One Planner & Journal App | React Native, Express.js, Firebase, Figma

bubblystudios.com/lively

- Created Lively, a React Native productivity app with dynamic calendars, mood tracking, live chat, and social event planning, scaling to 1K+ iOS App Store downloads and 10K+ monthly interactions
- Implemented real-time notifications and friend synchronization via Firebase on an Express.js server hosted on Render, and built custom UI components (infinite pager, bottom sheet) with React Native Gesture Handler and Reanimated

TECHNICAL SKILLS & AWARDS

Languages: Python, C++, Java, SQL, JavaScript, Swift, HTML/CSS

Frameworks & Technologies: PyTorch, TensorFlow, Scikit-Learn, Keras, Pandas, NumPy, Matplotlib, React, Git, Cursor

Certifications: Machine Learning & Intro to Statistical Learning (Stanford Online), CS50's Intro to Computer Science (HarvardX)

Honors & Awards: NCWIT Aspirations in Computing National Winner (Top 40 of 2.7K+ Applicants), National STEM Festival Finalist, U.S. Presidential Scholars Semifinalist, USACO Gold Division Competitor, 3x Congressional App Challenge Winner